

SQL Limit & Offset Notes

```
SELECT *  
FROM Employees  
LIMIT 10;
```

returns only 10 rows

Lowest 10 salaries

```
SELECT *  
FROM Employees  
ORDER BY Salary ASC  
LIMIT 10;
```

```
SELECT *  
FROM Employees  
LIMIT 5;
```

Is this wrong?

No.

Is it deterministic?

No.

The database does not guarantee which five rows you get.

Why is this dangerous?

Storage engines may

- change execution plans
- rebuild indexes
- insert new rows
- reorganize pages

As a result,

```
LIMIT 5
```

may return different rows over time.

SQL Dialects

MySQL	<pre>SELECT * FROM Employees LIMIT 5;</pre>
PostgreSQL	<pre>SELECT * FROM Employees LIMIT 5;</pre>
SQL Server	<pre>SELECT TOP 5 * FROM Employees;</pre>
Oracle	<pre>SELECT * FROM Employees FETCH FIRST 5 ROWS ONLY;</pre>

OFFSET

OFFSET tells SQL

Skip the first N rows, then return the remaining rows.

It is commonly used for pagination.

```
SELECT *  
FROM Employees  
LIMIT 5  
OFFSET 10;
```

Meaning

Skip 10 rows

Then return 5 rows

MySQL Shortcut

Equivalent syntax

```
SELECT *  
FROM Employees  
LIMIT 10,5;
```

Pagination Basics

Suppose a website displays 10 employees per page.

Page 1	LIMIT 10 OFFSET 0	Rows 1-10
Page 2	LIMIT 10 OFFSET 10	Rows 11-20
Page 3	LIMIT 10 OFFSET 20	Rows 21-30

Formula

$$\text{OFFSET} = (\text{Page} - 1) \times \text{PageSize}$$

Keyset Pagination

Instead of

```
OFFSET 100000
```

they use

```
WHERE ID > last_seen_id  
ORDER BY ID  
LIMIT 20;
```

This is called keyset pagination.